MILITARY SPECIFICATION

MOTOR, CONTROL

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification describes the requirements for the procurement of four types of Control Motor. The Control Motor is a two-phase induction motor having a high torque-to-inertia ratio, and approximately linear relationships between speed, control voltage, and torque. The control field of Alternating Motor types I, II and III consists of two independent windings which can be driven by a push-pull transistor amplifier, and an integral mechanical damper is incorporated in Control Motor types I and III. All four Control Motor types are for 400-cycle operation.

1.2 Classification. - The Control Motor shall be of the following types:

Type I	BUWEPS Drawing No. 1876583,	Motor, Control, Inertially Damped
Type II	BUWEPS Drawing No. 1877843,	Motor, Control
Type III	BUWEPS Drawing No. 1884648,	Motor-Control Inertially Damped
Type IV	BUWEPS Drawing No. 1981483	Motor, Control

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, form a part of this specification:

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SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Fiber
PPP-B-676	Boxes, Setup, Paperboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-L-10547	Liners, Case, Waterproof
MIL-P-16298	Preservation, Packaging, Packing and Marking of Electronic Machines having Rotating Parts and Associated Repair Parts
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

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STANDARDS

MILITARY

MIL-STD-106	Mathematical Symbols
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUWEPS Drawing No. 1876583, Motor, Control, Inertially Damped

BUWEPS Drawing No. 1877843, Motor, Control

BUWEPS Drawing No. 1884648, Motor-Control Inertially Damped

BUWEPS Drawing No. 1981483, Motor, Control

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements

(Copies of specifications, standards, and drawings requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

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AMERICAN GEAR MANUFACTURERS ASSOCIATION PUBLICATION

AGMA 207.04 — 20 Degree Involute Fine-Pitch System for Spur and Helical Gears

(Application for copies should be addressed to American Gear Manufacturing Association, 1 Thomas Circle, N.W., Washington 5, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 General Requirements

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the Control Motor shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the contractor, prior to his starting production, is capable of producing items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.2.

3.1.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the Control Motor shall conform to the detailed requirements given on the applicable BUWEPS drawing, in the specifications listed thereon, and in Specification MIL-F-18870.

3.1.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for the Control Motor shall be in accordance with the applicable BUWEPS drawing, and with the specifications and publications listed thereon.

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3.1.4 Moisture, fungus, and corrosion. - The Control Motor shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the applicable BUWEPS drawing, and in the specifications listed thereon.

3.1.5 Marking. - Marking of assemblies shall be in accordance with the applicable BUWEPS drawing.

3.1.6 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.1.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NO-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.1.8 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.9 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

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3.2 Definitions. - Definitions applicable to this specification and associated drawings are as follows:

3.2.1 Single phasing. - The tendency of the rotor to continue to rotate when one motor field is opened while the other motor field remains energized.

3.2.2 Starting voltage. - The maximum control field voltage necessary to start smooth, continuous shaft rotation (with no mechanical load) with rated voltage of proper phase and frequency applied to the fixed field when the control field voltage is increased slowly from zero.

3.2.3 No-load speed. - The speed of shaft rotation (with no mechanical load) with rated voltage of proper frequency and phase relationship applied to both motor fields.

3.2.4 Stall torque. - The torque developed at speeds in excess of one revolution per minute, but less than one-half percent of synchronous speed with rated voltage of proper frequency and phase relationship applied to both fields.

3.2.5 Ambient temperature. - The surrounding temperature of the unit along the shaft axis 1.5 plus or minus 1.0 inches away from the end opposite the shaft.

3.2.6 Temperature rise. - Winding temperature rise (degrees centigrade) is given by:

$$(234.5 + T_C)\,\frac{R_H - R_C}{R_C}$$

where R_H = resistance of windings at temperature T_H,
R_C = resistance of windings at temperature T_C,
T_H = temperature of windings when energized,
T_C = ambient temperature,

and all temperatures are in degrees centigrade.

Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

3.2.7 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperatures, and spore exposure, which present an opportunity for fungi growth.

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3.3 Performance requirements and product characteristics. - The Control Motor shall meet the following performance requirements and product characteristics.

3.3.1 Mechanical requirements

3.3.1.1 End play. - When measured in accordance with 4.4.4.1, the end play of the shaft shall not exceed 0.005 inch total indicator reading. See 4.4.4.1.

3.3.1.2 Radial play. - When measured in accordance with 4.4.4.2, the radial play of the shaft shall not exceed 0.001 inch total indicator reading. See 4.4.4.2.

3.3.1.3 Perpendicularity. - When measured in accordance with 4.4.4.3, the axial run-out of each mounting face shall not exceed 0.001 inch total indicator reading per inch of mounting face diameter. See 4.4.4.3.

3.3.1.4 Pinion. - The pinion gear teeth shall conform to AGMA 207.04. See 4.4.4.4.

3.3.1.4.1 Pitch diameter run-out. - When measured in accordance with 4.4.4.4.1, the pitch diameter run-out shall meet the requirements of table I. See 4.4.4.4.1.

3.3.1.4.2 Total composite error. - When measured in accordance with 4.4.4.4.2, the total composite error for each mounting diameter shall meet the requirements of table I. See 4.4.4.4.2.

3.3.1.4.3 Center distance. - When measured in accordance with 4.4.4.4.3, the center distance for each mounting diameter shall meet the requirements of table I. See 4.4.4.4.3.

3.3.2 Electrical requirements.

3.3.2.1 Dielectric strength and insulation resistance.

3.3.2.1.1 Dielectric strength. - When tested in accordance with 4.4.4.6.1, the Control Motor shall withstand the dielectric test voltage specified in 4.4.4.6.1 without damage, arcing, or breakdown. See 4.4.4.6.1.

3.3.2.1.2 Insulation resistance. - When measured in accordance with 4.4.4.6.2, the insulation resistance between winding and the housing and between each winding and each other winding shall meet the requirements of table I. See 4.4.4.6.2.

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3.3.2.2 Direction of rotation. - When checked in accordance with 4.5.4.7, the direction of shaft rotation shall be as specified on the applicable BUWEPS drawing. See 4.4.4.7.

3.3.2.3 Mechanical noise. - When checked in accordance with 4.4.4.8 the Control Motor shall be free from abnormal mechanical noise. See 4.4.4.8.

3.3.2.4 Single phasing. - When the Control Motor is tested in accordance with 4.4.4.8, the rotor shall come abruptly to a standstill from a no-load speed condition when either the control field or the fixed field is de-energized. See 4.4.4.9.

3.3.2.5 Starting voltage. - When measured in accordance with 4.4.4.10, the starting voltage for both directions of rotation shall meet the requirements of table I. See 4.4.4.10.

3.3.2.6 No-load speed. - When measured in accordance with 4.4.4.11, the no-load speed for both directions of rotation shall meet the requirements of table I. See 4.4.4.11.

3.3.2.7 Stall torque. - When measured in accordance with 4.4.4.12, the stall torque for both directions of rotation shall meet the requirements of table I. See 4.4.4.12.

3.3.2.8 Speed at one-half measured stall torque. - When measured in accordance with 4.4.4.13, the speed at one-half measured stall torque shall meet the requirements of table I. See 4.4.4.13.

3.3.2.9 Input current and power at stall. - When measured in accordance with 4.4.4.14, the input current and power to each field shall meet the requirements of table I. See 4.4.4.14.

3.4 Environmental requirements. - The Control Motor shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1 Vibration

3.4.1.1 Vibration, transportation. - The Control Motor shall be capable of withstanding the vibration tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.1.2 Vibration, operating condition. - The Control Motor shall withstand the type I vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.1.2.

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3.4.2 Shock

3.4.2.1 Shock, transportation, storage and handling. - The Control Motor shall be capable of withstanding the shock tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.2.2 Shock, non-operating condition. - The Control Motor shall withstand the type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.2.2.

3.4.3 Temperature

3.4.3.1 Temperature, operating condition. - The Control Motor shall withstand operational ambient temperatures of between minus 4 degrees F (minus 20 degrees C) and 185 degrees F (85 degrees C) in accordance with Specification MIL-F-18870, shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.1.

3.4.3.2 Temperature, non-operating condition. - The Control Motor shall withstand storage temperatures of between minus 80 degrees F (minus 62 degrees C) and 185 degrees F (85 degrees C) in accordance with Condition II of OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.3.2.

3.4.4 Humidity. - The Control Motor shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. See 4.5.4.

3.4.5 Tropical environment. - The Control Motor shall withstand the tropical environmental conditions specified in Specification MIL-E-5272 under "Fungus Resistance Tests", and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification. See 4.5.5.

3.4.6 Pinion Hardness. - When measured in accordance with 4.5.6, the hardness of the pinion shall be 28 minimum on the Rockwell C scale. See 4.5.6.

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3.4.7 Temperature rise at stall. - When measured in accordance with 4.5.7, the temperature rise at stall shall meet the requirements of table I. See 4.5.7.

3.4.8 Operating life. - The Control Motor shall withstand the operating life test described in 4.5.8, and thereafter shall have a no-load speed greater than 90 percent of the lowest no-load speed which satisfies the requirements of 3.3.2.6, and shall meet all the performance requirements and product characteristics of this specification with the exception that failure to comply with the requirements of 3.3.2.6 shall not be considered cause for rejection. See 4.5.8.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities acceptable to the Government. The Government preserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three Control Motors shall be manufactured by the methods and procedures proposed for the production lost. These shall be submitted to the Government inspector for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract,

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specification, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic production samples. - The tests of 4.4 and 4.5 shall be performed in steps as specified below. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory compliance of each unit from (3) three consecutive groups, select one (1) unit from each group of twenty-five (25).

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.

4.2 Classification of test. - The inspection and testing of the Control Motor shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each Control Motor submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.1 and 4.1.2. These tests are detailed in 4.5. Acceptance shall be based on no defects in the sample.

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4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the Control Motor. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 End play tester. - An end play tester or bench lathe capable of securing the Control Motor in a vertical or horizontal position and mounting a dial indicator.

4.3.2 Center distance tester. - A test fixture capable of securing the Control Motor in a vertical position, mounting a dial indicator, and measuring center distance to any accuracy of plus or minus 0.0002 inch. The standard inspection master gear of the center distance tester shall be that specified on the applicable BUWEPS drawing, and a constant force of 8 ounces shall be maintained between the pinion and the standard inspection master gear.

4.3.3 Dial indicator. - A dial indicator which can be mounted on the testers of 4.3.1 and 4.3.2, and which measures with an accuracy of plus or minus 0.0001 inch.

4.3.4 Mechanical loads.

4.3.4.1 Reversing 1-pound load. - A reversing 1-pound load which can be applied axially to the shaft of the Control Motor.

4.3.4.2 Reversing 1/2-pound load. - A reversing 1/2-pound load which can be applied radially to the shaft of the Control Motor.

4.3.4.3 Dynamometer. - A dynamometer capable of applying a maximum 5 ounce-inch torque, with a dial calibrated to 0.1 ounch-inch and an accuracy of plus or minus 5 percent. The dynamometer shall be capable of speeds up to 10,000 revolutions per minute.

4.3.5 Heat sinks.

4.3.5.1 Control Motor type I. - The heat sink shown in figure 1 for Control Motor type I.

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4.3.5.2 Control Motor types II, III, and IV. - The heat sink shown in figure 1 for Control Motor types II, III, and IV.

4.3.6 High-potential tester. - A high-potential tester capable of applying up to 1000 volts root-mean-square (R.M.S.), 60 cycles per second. The rate of change of output voltage shall not exceed 75 volts per second when the unit is turned on or off.

4.3.7 Insulation resistance tester. - An insulation resistance tester capable of applying 500 volts direct current, and measuring resistance up to 100 megohms.

4.3.8 Power sources.

4.3.8.1 400-cycle motor power supply. - A 400-cycle, two-phase power supply having the following characteristics:

(a) Volt-amperes per phase. - 25 volt-amperes minimum,

(b) Volt per phase. - 115 volts plus or minus 1 percent for one phase and continuously adjustable from 0 to 130 volts for the other phase,

(c) Phase angle between voltages of fundamental frequency. - 90 degrees plus or minus 5 degrees,

(d) Frequency. - 400 plus or minus 20 cycles per second,

(Waveform. - Sinusoidal, with harmonic content not greater than 5 percent of the voltage of fundamental frequency.)

4.3.8.2 60-cycle equipment power supply. - A 60 cycle power source capable of supplying adequate volt-amperes to any 60-cycle test equipment used in conducting the tests of this specification. The voltage shall be 115 plus or minus 2 volts, the frequency 60 plus or minus 3 cycles per second, and the waveform shall be sinusoidal with harmonic content not greater than 5 percent of the voltage of fundamental frequency.

4.3.9 Meters.

4.3.9.1 Voltmeter. - A multi-range, alternating-current voltmeter, 1, 3, 10, 30, 100, and 300 volts full scale, having an input impedance of at least 20,000 ohms per volt, and an accuracy of plus or minus 2 percent of full scale.

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4.3.9.2 Milliammeter. - A multi-range, alternating-current milliammeter, 250 and 500 milliamperes full scale, having an impedance not greater than 0.15 ohm, and an accuracy of plus or minus 2 percent of full scale.

4.3.9.3 Wattmeter. - A wattmeter capable of measuring up to 15 watts which can be used with voltages up to 130 volts and currents up to 0.5 ampere, and which has an accuracy of plus or minus 2 percent of full scale.

4.3.10 Capacitor. - A non-polarized capacitor having a capacitance of 0.25 microfarad plus or minus 10 percent, and a direct-current voltage rating of 200 volts minimum.

4.3.11 Stroboscope. - A stroboscope capable of flashing at rates up to 10,000 flashes per second, and having an accuracy of plus or minus 2 percent of indicated flashing rate.

4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the Control Motor is in compliance with the requirements of 3.1, 3.3, 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

4.4.2 Test conditions. - Unless otherwise specified, the Control Motor shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees F (18 to 35 degrees C).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

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4.4.4.1 End play. - With the Control Motor mounted horizontally in the tester of 4.3.1, the housing held stationary, and the indicator of 4.3.3 mounted to measure axial movement of the shaft, the reversing 1-pound load of 4.3.4.1 shall be applied axially to the shaft, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.1.

4.4.4.2 Radial play. - With the Control Motor mounted horizontally in the tester of 4.3.1, the housing held stationary, and the indicator of 4.3.3 mounted to measure radial movement of the shaft as close as possible to the face of the bearing, the reversing 1/2-pound load of 4.3.4.2 shall be applied radially to the shaft 1/4 inch maximum from the face of the bearing, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.2.

4.4.4.3 Perpendicularity. - The following procedure shall be followed for each mounting face. With the Control Motor mounted vertically in the tester of 4.3.1, the shaft held stationary, and the indicator of 4.3.3 mounted to measure axial movement of the mounting face not more than 0.062 inche from its outside diameter, the housing shall be rotated by hand, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.3.

4.4.4.4 Pinion. - The following procedure shall be followed to determine compliance with the requirements of 3.3.1.4. The tests of 4.4.4.4.1, 4.4.4.4.2, and 4.4.4.4.3 need not be performed separately, as the test of 4.4.4.4.3 supplies the data required for 4.4.4.5.1, 4.4.4.4.2, and 4.4.4.4.3.

4.4.4.4.1 Pitch diameter run-out. - With the Control Motor mounted vertically in the tester of 4.3.2, the housing held stationary, and the indicator of 4.3.3 mounted to measure radial movement of the standard inspection master gear, the pinion shall be rotated by hand, and the total indicator reading shall be observed to determine compliance with the requirements of 3.3.1.4.1.

4.4.4.4.2 Total composite error. - The following procedure shall be followed for each mounting diameter. With the Control Motor mounted vertically in the tester of 4.3.2, the housing held stationary, and the indicator of 4.3.3 mounted to measure radial movement of the standard inspection master gear, the maximum and minimum indicator readings obtained as the shaft is rotated by hand shall be observed for three different positions of the housing 120 degrees apart, and the difference between the smallest and the largest of the six indicator readings obtained shall be noted to determine compliance with the requirements of 3.3.1.4.2.

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4.4.4.4.3 Center distance. - The following procedure shall be followed for each mounting diameter. With the Control Motor mounted vertically in the tester of 4.3.2, the housing held stationary, and the indicator of 4.3.3 mounted to measure radial movement of the standard inspection master gear, maximum and minimum center distances obtained as the shaft is rotated by hand shall be observed for three different positions of the housing 120 degrees apart, and the smallest and the largest of the six center distance readings shall be noted to determine compliance with the requirements of 3.3.1.4.3.

Note 1. - For the tests of 4.4.4.6 through 4.4.4.13, the following conditions shall obtain

1. Mounting.

a. Control Motor type I shall be mounted on the heat sink of 4.3.5.1,

b. Control Motor types II, III, and IV shall be mounted on the heat sink of 4.3.5.2,

c. The heat sink shall be mounted on a heat-insulating base,

d. The Control Motor shall be shielded from room air currents.

2. Environmental conditions.

a. Ambient temperature. - 68 to 86 degrees F (20 to 30 degrees C),

b. Atmospheric pressure. - 30 plus or minus 2 inches of mercury,

c. Vibration. - None,

d. Relative humidity. - 75 percent maximum.

3. Warm-up. - The Control Motor shall be given a warm-up period of one hour with rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field, using the power supply of 4.3.8.1.

4. Application of control field voltage. - Except for the tests of 4.5.10 and 4.5.11, control field voltage shall be applied as briefly as possible to avoid increasing the temperature of Control Motor from its temperature at the end of the warm-up period.

4.4.4.6 Dielectric strength and insulation resistance.

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4.4.4.6.1 Dielectric strength. - Using the high-potential tester of 4.3.6, the following voltages shall be applied for 15 plus or minus 5 seconds (with no other connections made to the Control Motor) while observing the Control Motor to determine compliance with the requirements of 3.3.2.1.1. The high-potential voltages specified below are for testing by the vendor of the Control Motor. Testing by users of the Control Motor shall be at 80 percent of these voltages. Excessive high potential testing should be avoided.

a. Winding to housing. - 500 volts r.m.s. for windings rated at 60 volts r.m.s. or less, and 1000 volts for windings rated at greater than 60 volts r.m.s. (this includes the isolated halves of the control field of Control Motor type II),

b. Winding to winding. - 500 volts r.m.s. when both windings are rated at 60 volts r.m.s. or less, and 1000 volts r.m.s. when either or both windings are rated at greater than 60 volts r.m.s.

4.4.4.6.2 Insulation resistance. - Immediately after application of high-potential voltage in accordance with 4.4.4.6.1, the insulation resistance shall be measured between each winding and every other winding and between each winding and the housing, using the insulation resistance tester of 4.3.7, to determine compliance with the requirements of 3.3.2.1.2.

4.4.4.7 Direction of rotation. - With the Control Motor connected in accordance with the circuit for checking direction of rotation shown on the applicable BUWEPS drawing, and using the adjustable voltage phase of the power supply of 4.3.8.1, and the capacitor of 4.3.10, the direction of shaft rotation shall be noted to determine compliance with the requirements of 3.3.2.2.

4.4.4.8 Mechanical noise. - With the Control Motor connected as shown in figure 2, and rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, the unit shall be checked for freedom from abnormal mechanical noise such as knocking or intermittent scraping sounds for both directions of rotation in the horizontal, vertical-shaft up, and vertical-shaft down positions to determine compliance with the requirements of 3.3.2.3.

4.4.4.9 Single phasing. - With the Control Motor connected as shown in figure 2, and rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, switch S_1 shall be opened while observing the pinion after the unit has attained full no-load speed. This procedure shall be repeated for the otherposition of switch S_3, and for switch S_2 opened instead of S_1 (for both positions of switch S_3), to determine compliance with the requirements of 3.3.2.4.

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4.4.4.10 Starting voltage. - With the Control Motor connected in accordance with figure 2, and rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field, the voltage applied to the control field shall be increased slowly from zero until the pinion rotates continuously, and the control field voltage corresponding to this condition shall be noted. This test shall be performed at least three times for each direction of rotation, and the greatest value of control field voltage thus measured shall be noted to determine compliance with the requirements of 3.3.2.5.

4.4.4.11 No-load speed. - With the Control Motor connected in accordance with figure 2, and rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, the no-load speed in each direction of rotation shall be measured, using the stroboscope of 4.3.11, to determine compliance with the requirements of 3.3.2.6.

4.4.4.12 Stall torque. - With the Control Motor connected in accordance with figure 2, and rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, the stall torque in each direction of rotation shall be measured, using the dynamometer of 4.3.4.3, to determine compliance with the requirements of 3.3.2.7.

4.4.4.13 Speed at one-half measured stall torque. - With the Control Motor connected in accordance with figure 2, rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, and a torque load equal to one-half the stall torque measured in 4.3.4.12 applied using the dynamometer of 4.3.4.3, the speed in both directions of rotation shall be measured using the stroboscope of 4.3.11 to determine compliance with the requirements of 3.3.2.8.

4.4.4.14 Input current and power at stall. - With the Control Motor connected in accordance with figure 2, rated voltage as specified on the applicable BUWEPS drawing applied to the fixed field and control field, and the pinion locked by a clamp, the current and power to each field shall be measured, using the milliammeter of 4.3.9.2 and the wattmeter of 4.3.9.3, to determine compliance with the requirements of 3.3.2.9.

4.4.4.15 Preservation, packing and packaging. - Preservation, packing and packaging of the Control Motor shall conform to section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Vibration.

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4.5.1.1 Vibration, transportation. - The Control Motor, packaged and packed as specified in section 5, shall be vibration tested as specified in Specification MIL-W-21927.

4.5.1.2 Vibration, operating condition. - The Control Motor, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.4.1, and shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.5.2 Shock.

4.5.2.1 Shock, transportation, storage and handling. - The Control Motor, packaged and packed as defined in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.2.2 Shock, non-operating condition. - The Control Motor, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in 3.4.3, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3 Temperature.

4.5.3.1 Temperature, operating condition. - The Control Motor shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.3.2 Temperature, non-operating condition. - The Control Motor shall be subjected to temperature tests so as to meet the requirements specified in 3.4.3.2 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.4 Humidity. - The Control Motor shall be subjected to humidity tests so as to meet the requirements specified in 3.4.4, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.5 Tropical environment. - The Control Motor shall be exposed to fungus tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

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4.5.6 Pinion hardness. - The hardness of the pinion shall be measured by marking an indentation on the flat end surface of the pinion, near its center, using a suitable hardness testing machine, and measuring the size of the indentation to determine compliance with the requirements of 3.4.6.

Note 2. - For the tests of 4.5.7 and 4.5.8, the conditions listed in note 1 shall obtain.

4.5.7 Temperature rise at stall. - With the Control Motor connected in accordance with figure 2, rated voltages as specified on the applicable BUWEPS drawing applied to the fixed field and control field, and the pinion locked by a heat-insulating clamp, the unit shall remain energized until the change of winding resistance over a five-minute period is less than 0.4 percent of winding nominal resistance at room temperature. When this condition is reached, the temperature rise shall be computed using the formula of 3.2.6 to determine compliance with the requirements of 3.4.7.

4.5.8 Operating life. - With the Control Motor connected in accordance with figure 2, no torque load, and rated voltage applied to the fixed field and control field in accordance with the applicable BUWEPS drawing, the Control Motor shall be operated 400 hours in the vertical - shaft-up position, 300 hours in the vertical - shaft-down position, and then 300 hours in the horizontal position, for a total operating period of 1000 hours. At the end of the 1000-hour operating period, the no-load speed shall be measured in accordance with 4.4.4.11 to determine compliance with the no-load speed requirements of 3.4.8.

4.6 Waiver of tests. - Tests specified in 4.5.1.1 and 4.5.2.1 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. - At the conclusion of the tests specified in 4.5.1.1 and 4.5.2.1, functional and other tests, as applicable, shall be conducted on the Control Motor to determine freedom from operational malfunction and compliance with all the requirements of section 3.

4.8 Failure of the Control Motor to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-16298.

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5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

5.1.1.2 Unit packaging. - Unit packaging of the motor shall be in accordance with Method IIe of Specification MIL-P-116. The unit package shall contain one (1) motor packaged in a paperboard box in accordance with Specification PPP-B-566 or PPP-B-676 with Kraft paper overwrap in accordance with Specification UU-P-268.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the motor will meet all the requirements of Section 3 after being tested in accordance with 4.5.1.1 and 4.5.2.1.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5%. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit-packaging. - Unit packaging of the motor shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one (1) motor.

5.1.3.3 Intermediate package. - Not applicable.

5.2 Packing.

5.2.1 Level A. - In addition the following, Level A packing shall be in accordance with Specification MIL-P-16298.

5.2.1.1 Exterior containers. - Motors packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, Class 2 (gross weight not to exceed 150 lbs).

5.2.1.2 Case liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

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5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-16298.

5.2.2.1 Exterior containers. - Motors packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type or PPP-B-621, Class 1 (gross weight not to exceed 150 lbs) or in fiber boxes in accordance with Specification PPP-B-636, Class 1 (gross weight not to exceed 70 lbs).

5.2.2.2 Case liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

5.2.3 Level C. - Motors packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. - None, unless otherwise specified.

5.3.2 Normal markings. - In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. - The Control Motor is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiver Sets AN/BQG-1, -2, -3, -4, or other suitable application.

6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels of packaging and packing; number of unit containers per pack; and guarantee clause (see 6.4).

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to para. 4.6.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

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For level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For level B. - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any Control Motor which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.5 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

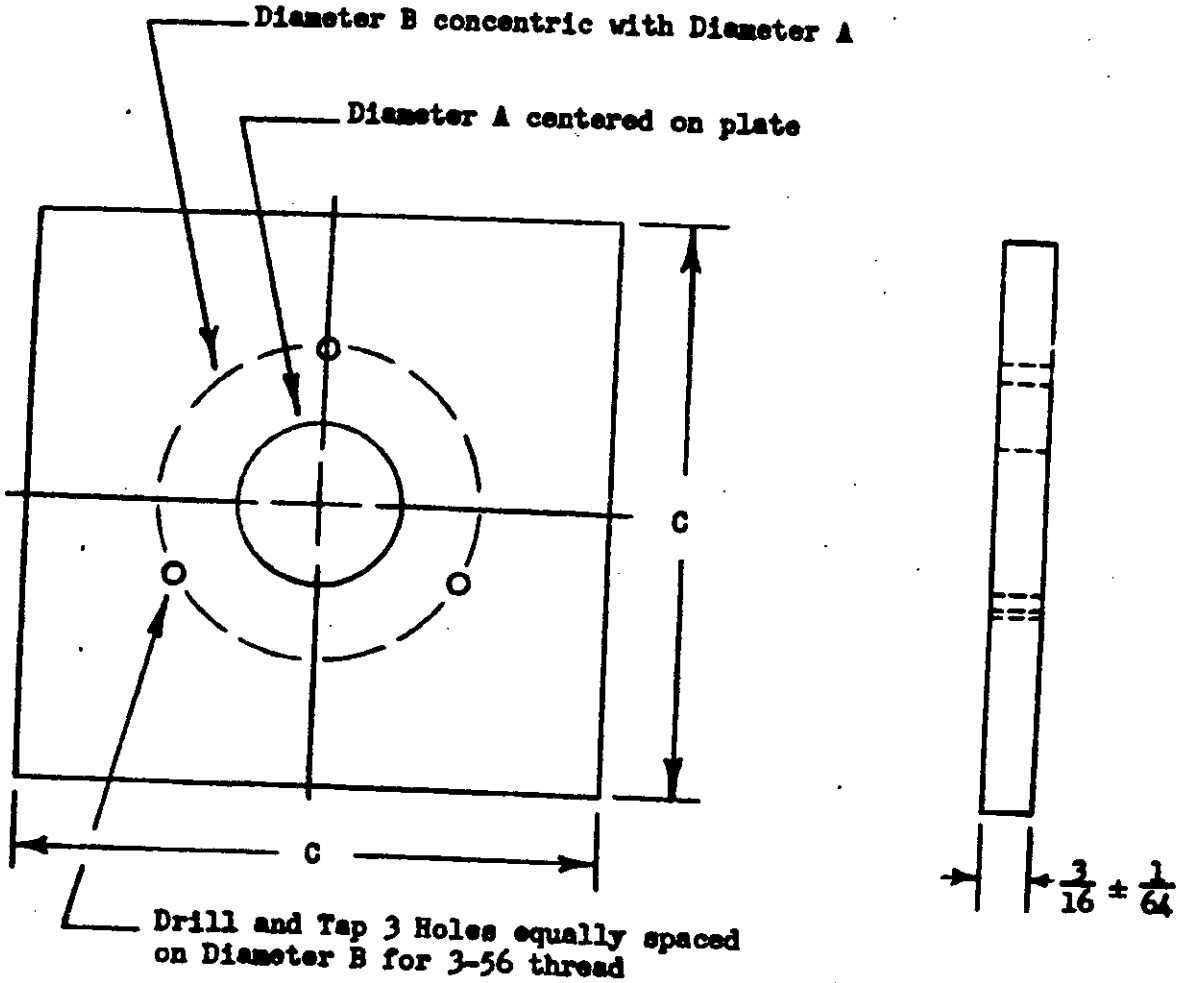
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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TABLE I. CONTROL MOTOR REQUIREMENTS

Quantity	Control Motor Type			
	I	II	III	IV
Pinion				
Pitch Diameter Run-Out (Max)	0.001 T.I.R.	0.001 T.I.R.	0.001 T.I.R.	0.001 T.I.R.
Total Composite Error (Max)	0.0025 T.I.R.	0.002 T.I.R.	0.002 T.I.R.	0.002 T.I.R.
Center Distance	0.828 $^{+.0000}_{-.0025}$	0.554 $^{+.000}_{-.002}$	0.554 $^{+.000}_{-.002}$	0.554 $^{+.000}_{-.002}$
Insulation Resistance (Min)	25 Megohms	10 Megohms	25 Megohms	10 Megohms
Starting Voltage (Max)	1 V.	3 V.	1 V.	12 V.
No-Load Speed (Min)	4800 R.P.M.	6200 R.P.M.	6200 R.P.M.	6200 R.P.M.
Stall Torque (Min)	1.30 Oz.-In.	0.55 Oz.-In.	0.55 Oz.-In.	1.35 Oz.-In.
Speed at One-Half Measured Stall Torque (Max)	3600 R.P.M.	4800 R.P.M.	4800 R.P.M.	4800 R.P.M.
Input Current at Stall				
Fixed Field (Max)	0.120 A.	0.066 A.	0.066 A.	0.210 A.
Control Field (Max)	0.340 A.	0.066 A.	0.176 A.	0.210 A.
Input Power at Stall				
Fixed Field (Max)	6.8 W.	3.9 W.	3.9 W.	10 W.
Control Field (Max)	6.8 W.	3.9 W.	3.9 W.	10 W.
Temperature Rise at Stall (Max)	65° C.	65° C.	65° C.	100° C.

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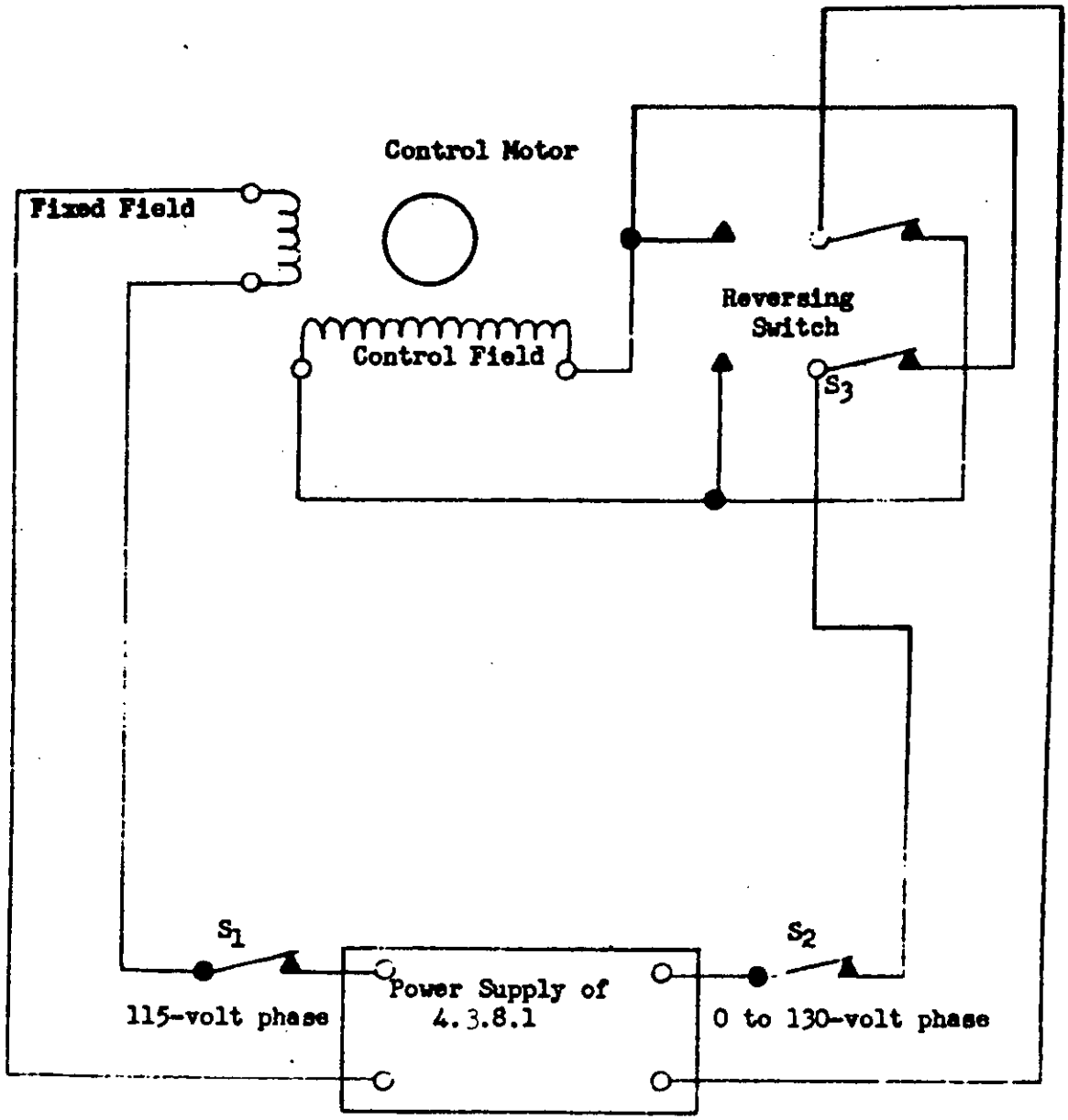


Material: Aluminum, flat sheet, temper T3, Specification QQ-A-355
Finish: Black Anodized

Control Motor Type	+.001 A -.000 Inch	+.002 B -.002 Inches	+.050 C -.050 Inches
I	0.876	1.688	4.30
II, III, IV	0.626	1.375	3.20

Figure 1. Control Motor Heat Sinks

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Note. When the control field consists of two windings, the windings shall be connected in series in accordance with the circuit for checking direction of rotation given on the applicable BUORD drawing.

Figure 2. Test Circuit for Control Motor

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